

ProbOS — Month 3 Plan

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July 5, 2026

Status

Per `docs/monthly_plans/overall/main.tex`, Month 3 was always DIRECTIONAL rather than pre-specified in detail – Month 1’s retrospective explicitly flagged over-specifying future months as a mistake, corrected by deferring detailed week-by-week planning until the month is actually about to begin. This document intentionally states THEMES, not a committed day-by-day schedule; the latter will be written (following the same plan-then-retrospective process used for every week so far) once Month 3’s actual work begins.

Candidate themes, in rough priority order

GPU kernel path

Referenced as a genuine open item in `docs/sbir_phase1_onepager.md`’s SBIR Ask (“GPU kernel, 10x speedup over CPU, building on the existing C++/OpenMP path”). The Month 1 Week 4 C++/OpenMP kernel achieved a measured 7x speedup over the Python engine; a GPU path would need its own honest profiling discipline (the same “measure, don’t assume” standard that found DP’s real 1040x/45x wins alongside one genuine non-win in Month 1 Week 4) before any speedup number is claimed.

Extend the REST API’s model registry – DONE (Week 9)

Completed: `python/server/main.py` now registers all four models (battery, option pricer, ED queue, clinical trial), with `/filter` correctly and honestly rejecting `clinical_trial` rather than returning a meaningless result. See `docs/monthly_plans/month3/week9/main.tex` for the full retrospective.

Pharmaceutical stability model – NEW, near-term priority

Per the strategic reassessment documented in `docs/vision/main.tex`: drug degradation kinetics are Arrhenius reaction kinetics – the exact same mathematical structure as `BatteryModel2Cell`’s SEI decomposition, with drug potency as the depleting quantity in place of battery capacity. An estimated 80% of the existing model architecture transfers directly; remaining work is domain-appropriate parameter priors and validation against a real stability-study dataset, matching the same discipline used for Kim (2007). This directly serves FDA CMC / ICH Q1A requirements, the same regulatory pattern already validated for battery safety.

Automotive / EV safety positioning – NEW, near-term priority

Not new engineering – the existing, already-validated battery model, reframed for ISO 26262’s hazard-analysis and documented- failure-mode requirements, which are structurally analogous to what FDA/NRC already require. The primary remaining work is outreach and positioning specific to automotive/EV safety engineers, not new model development.

PDSL v0.2 – control flow

Month 1 Week 4’s PDSL compiler v0.1 supports only flat drift/diffusion declarations, no loops or conditionals. Genuinely useful, but not urgent – no current model has needed it yet.

Apply the particle filter to real (non-synthetic) data

The most consequential open SBIR ask (“apply the particle filter to real sensor data from a pilot partner”), and the one most dependent on external factors (an actual pilot partnership) rather than purely technical work. Cannot be scheduled with certainty; flagged here so it is not forgotten.

Continued quality-standard maintenance

Per the mid-Month-2 hardening pass: `pre_week_audit.sh` and `docs/standards/quality_standards.md` are living documents. Any new gap discovered (declared-but-unenforced checks, stale documentation, vestigial code patterns) should be closed in the same session it is found, not deferred – the standing practice established this month, to be continued.

Process

As with every prior month, Month 3’s actual week-by-week plan will be written and compiled to PDF (`docs/monthly_plans/month3/weekN/main.tex`) before that week’s work begins, with a retrospective appended once it completes. This document will be revisited and made more concrete once Month 3 actually starts.